

A List of Unnamed Probability Distributions

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2026-09-08

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List of PMF

1.
$$P(x) = \frac{x-3}{6}, \quad \text{for } x = 4, 5, 6$$
2.
$$P(x) = \frac{3x+4}{21}, \quad x = 0, 1, 2$$
3.
$$P(X) = \frac{3-|4-x|}{k}; \quad x = 2, 3, 4, 5, 6$$
4.
$$P(X) = \frac{5-|6-x|}{k}; \quad x = 1, 2, 3, 4, 5$$
5.
$$P(X) = \frac{7-|5-x|}{k}; \quad x = 1, 2, 3, 4, 5, 6$$
6.
$$P(X) = \frac{6-|7-x|}{k}; \quad x = 1, 2, \dots, 10$$

7.

$$P(x) = \frac{x+1}{k}; \quad x = 1, 2, 3, 4$$

8.

$$P(x) = \frac{2x+k}{56}; \quad x = -3, -2, -1, 0, 1, 2, 3$$

9.

$$P(x) = \frac{1}{14}(a+2x); \quad x = -3, -2, -1, 0, 1, 2, 3$$

10.

$$P(x) = k(x-2); \quad x = 3, 4, 5, 6, 7, 8$$

11.

$$P(x) = \frac{x-1}{k}; \quad x = 2, 3, 4, 5$$

12.

$$p(x) = \frac{x+4}{30}; \quad x = 0, 1, 2, 3, 4$$

13.

$$P(x) = \frac{2x+1}{9}; \quad x = 0, 1, 2$$

Is it a PMF?

Joint PMF

1.

$$P(X, Y) = \frac{x+2y}{28}; \quad x = 0, 1; \quad y = 0, 1, 2, 3$$

2.

$$P(X, Y) = \frac{2x+3y}{45}; \quad x = 0, 1, 2; \quad y = 0, 1, 2$$

3.

$$P(X, Y) = \frac{x+y+1}{42}; \quad x = 0, 1, 2; \quad y = 0, 1, 2, 3$$

4.

$$P(X, Y) = \frac{2x+y+1}{52}; \quad x = 1, 2; \quad y = 1, 2, 3, 4$$

5.

$$P(X, Y) = \frac{3x+y}{48}; \quad x = 1, 2; \quad y = 0, 1, 2, 3$$

6.

$$P(x, y) = \frac{1}{21}(x + y); \quad x = 1, 2, 3; \quad y = 1, 2$$

7.

$$P(x, y) = \frac{x + y}{9}; \quad x = 0, 1, 2; \quad y = 0, 1$$

PDF

1.

$$f(x) = 6x(1 - x); \quad 0 \leq x \leq 1$$

2.

$$f(z) = \frac{2}{9}(3z - z^2); \quad 0 \leq z \leq 3$$

3.

$$f(x) = k(x + 1); \quad 0 \leq x \leq 1$$

4.

$$f(x) = kx(x - 1); \quad 1 \leq x \leq 4$$

5.

$$f(x) = kx^2 + kx + \frac{1}{8}; \quad 0 \leq x \leq 2$$

6.

$$f(x) = \frac{1}{30}(k + 2x); \quad 2 < x < 5$$

7.

$$f(x) = 2x; \quad 0 < x < 1$$

8.

$$f(x) = \frac{1}{30}(3 + 2x); \quad 2 < x < 5$$

9.

$$f(x) = ax^2; \quad 0 < x < 4$$

10.

$$f(x) = kx; \quad 0 < x < 4$$

11.

$$f(x) = 3x^2; \quad 0 \leq x \leq 1$$

12.

$$f(y) = k(3y + 5); \quad 1 < y < 5$$

13.

$$f(x) = \frac{1}{b-a}; \quad a \leq x \leq b$$